

Goeppert 2D MoS2 Parabolic Flight Test

STATEMENT OF WORK

The development of advanced semiconductor manufacturing techniques and materials aligned with the priorities of the “CHIPS for America” (<https://www.nist.gov/chips>) of the CHIPS and Science Act of 2021 is crucial for maintaining United States technological leadership in both terrestrial and space applications that require high performance semiconductors. NASA’s In Space Production Applications (InSPA) Portfolio enables commercially viable and sustainable business opportunities in low-Earth orbit (LEO) that advance solutions for national interest or public benefit by leveraging the capabilities and resources of the ISS. Goeppert LLC recently completed a NASA SBIR Phase III contract to develop a ground prototype of a novel specialized vacuum furnace system designed to process 4-inch wafers of 2-Dimensional Molybdenum Disulfide (MoS2) semiconductor material simultaneously while maintaining precise temperature control and environmental conditions during microgravity periods. The following describes the manner in which Goeppert LLC will develop and demonstrate a novel microgravity-enhanced annealing process in parabolic flight to de-risk critical technologies and procedures required for future low-Earth orbit (LEO) manufacturing for future demonstrations on the International Space Station (ISS) and other commercial spacecraft.

1. Objectives and Requirements

This project investigates how microgravity conditions during the critical annealing phase of MoS2 processing can enhance material quality and performance. Building upon preliminary research that suggests gravitational effects influence crystal formation and defect density, this effort will utilize parabolic flight testing to validate key process parameters and system capabilities. Through systematic investigation of microgravity effects on MoS2 annealing, this work will establish foundational data and operational experience necessary for future low Earth orbit manufacturing initiatives. The project will deliver both technical validation of the microgravity enhancement concept and practical operational knowledge required for eventual space-based production systems.

The main objectives of this Purchase Order are to:

- (1) develop the microgravity annealing system,
- (2) conduct parabolic flight validation of MoS2 annealing, and
- (3) perform materials characterization and analysis.

NASA will purchase parabolic flight services to enable microgravity testing of the Goeppert of MoS2 crystal manufacturing device. After the parabolic flight tests are accomplished, Goeppert shall conduct post-flight testing and analysis of MoS2 crystals and deliver flight and ground samples to the Government for independent testing and validation of results. Finally, Goeppert shall prepare and deliver a report on the results of microgravity annealing.

Results will be compiled into detailed reports correlating processing conditions with material quality improvements, providing quantitative assessment of microgravity enhancement effects and establishing baseline performance metrics for future space-based processing development. Key quality metrics to be evaluated include:

- Quantitative assessment of grain size enhancement,
- Detailed mapping and statistical analysis of adlayer formation,
- Comprehensive evaluation of sulfur vacancy density in microgravity-processed regions compared to control samples
- Correlation of material quality parameters with processing conditions, particularly focusing on the relationship between gravitational conditions and crystal formation characteristics

Goeppert shall provide NASA with flight and ground samples of MoS₂ crystals for use in independent test and evaluation to validate Goeppert results. Goeppert shall select paired flight/ground samples that are representative of the best quality produced.

a. Shipping Information:

[REDACTED]
NASA Langley Research Center

[REDACTED]

[REDACTED]

22 West Taylor Street
Hampton, VA 23681-2199

[REDACTED]

[REDACTED]

2. Specifications: The microgravity annealing system should maintaining stable processing conditions during parabolic flight, including vacuum levels of 7mPa and temperatures up to 800°C, including integrated temperature control, pressure monitoring, and real-time imaging capabilities with synchronized G-force measurement. The MoS₂ grain size enhancement should be greater than 50% increase in average domain size compared to 1g results, with greater than 30% reduction in adlayer formation.

3. Place of Performance: Goeppert LLC, Philadelphia PA.

4. Period of Performance: award thru February 2026

5. Schedule and Milestone Payments

The Estimated Completion Date (ECD) of milestones and payments shall be as follows:

Milestone	Description	Due Date	Amount of Payment (\$)
1	Kickoff Meeting and Zero-G Flight Reservation	Thirty (30) Days Following Award	[REDACTED]
2	Parabolic Design of Experiment Delivered to NASA	Ninety (90) Days Following Award	[REDACTED]
3	Safety Data Submitted to Zero-G Corp	Sixty (60) Days prior to Flight	[REDACTED]

4	Post-Flight Debrief to NASA	Two (2) weeks following flight	██████████
5	Final Report and Sample Delivery to NASA	Ninety (90) Days Following Parabolic Flight Completion	██████████

The total firm fixed price of this purchase order is ██████████